INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE
(Affiliated to UEM Kolkata and Approved by AICTE)

Degree:**B.C.A**, Stream:**B.C.A**

End Semester Examination 2024 - Second Semester Examination,April-2024

Paper Code: BCAAE201

Paper Name: Discrete Structure

Full Marks:100 Duration:3 Hours

Part - A

Attempt 10 questions

Each question carries 2 Marks (2 X 10)

1. Define Relation between two sets with examples. 2

or

1. Define Universal Relation with examples. 2

2. What is a power set? Explain with examples. 2

or

2. Define the term implication i.e. A---> B. 2

3. Define Deterministic finite automata (DFA). 2

or

3. Draw the Venn diagram of the complement of intersection of two sets. 2

4. Define adjacent edges in a graph. Draw a graph showing four adjacent edges. 2

or

4. Describe the concept of subgraph with proper examples. 2

5. Explain the properties of a complete graph. 2

or

5. Define Void relation with example. 2

6. Describe distributive law of propositional logic with example. 2

or

6. Can a regular graph with five vertices be drawn? Explain your answer with proper reason. 2

7. Explain when is a mapping called bijective mapping? 2

or

7. Find the number of elements in the power set of {a,b,c,d}. 2

8. Find the elements of the set S={x: x+7=7}. 2

or

8. Give example of a relation which is reflexive but not symmetric. 2

9. Define binary tree. 2

or

9. Draw a tree with exactly six vertices. 2

10. A graph (without any loop) has 8 vertices and 12 edges. Then what is the size of its adjacency matrix? 2

or

10. How can you define the order of an incidence matrix? 2

Part - B

Attempt 6 questions

Each question carries 5 Marks (5 X 6)

11. Let R be a relation in the set of integers Z defined by xRy if $(x \div y)$ is a multiple of 2 for all x,y belonging to Z. Analyze and Check whether R is symmetric and transitive relation. 5

or

11. If A={1,2}, B={2,3}, C={3,4}, then find 5

(i) (AxB)U(AxC) (ii) (AxB)$\cap$(AxC)

12. If A={1,2,3,4,5,6,7,8,9}, B={2,3,4,5}, C={2,4,6,8} and D={4,5,6,7} then 5

i. Verify the distributive law $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

ii. Verify De-Morgan's law : $A-(B \cup C) = (A-B) \cap (A-C)$

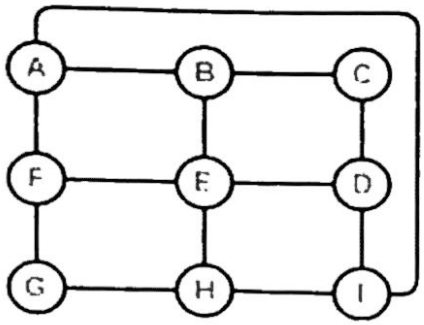
or

12. Show that the mapping f: R$\rightarrow$R defined by $f(x)=x^2$ is not injective. 5

13. Analyze the given values when A={1,2,3,4,5},B={1,2,4,8},C={1,2,3,5,7} and find out set which contains union of the set A and the symmetric difference of B and C. 5

or

13. Check whether the following graph is Hamiltonian or not: 5



14. Draw a graph with 9 vertices having degree 1,5,2,7,10,8. 5

or

14. Determine the adjacency matrix of the following graph: 5

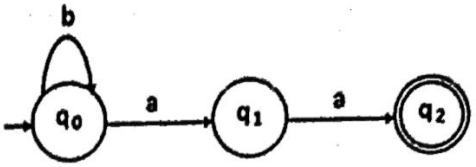


15. Analyze whether the following is a tautology or not. 5

(A ∧ B)-> (B ∨C)

or

15. Select the language accepted by the following DFA . 5



16. Show that the mapping f: N→N defined by f(x)=x+1 is not surjective. 5

or

16. Show that the mapping f: R→R defined by $f(x)=x^2$ is not surjective. 5

Part - C

Attempt 5 questions

Each question carries 10 Marks (10 X 5)

17. Show that the mapping f:R→R defined by f(x)=3x+5 is bijective. Determine f^{-1}. 10

or

17. Show that f: R-{3}→R-{1} given by f(x)= (x-2)/(x-3) is a bijective mapping. Also find its inverse. 10

18. Design a DFA that accepts all strings of {0, 1} starting with 01 then any no of 0 then 1 can exist. 10

or

18. Design a DFA that accepts language $a^m b^n c$ where n>=0. 10

19. In a survey of 320 persons, number of persons taking tea is 210, taking milk 100 and coffee is 70. Number of persons who take

tea and milk is 50, milk and coffee is 30, tea and coffee is 50. The number of persons taking all the three together is 20. Summarize the value which gives the number of persons who take neither tea nor milk nor coffee. 10

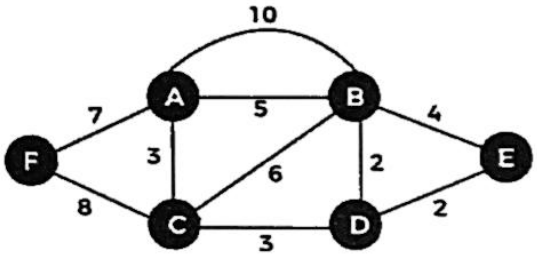
or

19. Draw a connected graph with 10 vertices 10

i. having 7 edges

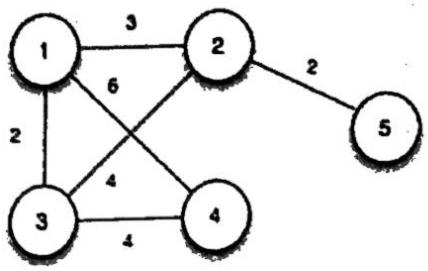
ii. having 50 edges

20. Find by Prim's Algorithm a minimal spanning tree from the following graph. Find an alternative spanning tree and compare it with the previous result. 10



or

20. Find by Prim's Algorithm a minimal spanning tree from the following graph 10



21. Analyse and show that intersection of two equivalence relations is also an equivalence relation. 10

or

21. Find by Kruskal's Algorithm a minimal spanning tree from the following graph 10

